



Management of IgA Vasculitis with Nephritis

Jean-Daniel Delbet^{1,2,3} · Cyrielle Parmentier^{1,2} · Claire Herbez Rea^{1,2} · Antoine Mouche^{1,2} · Tim Ulinski^{1,2}

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Abstract

Immunoglobulin A (IgA) vasculitis (IgAV), previously called Henoch–Schönlein purpura, is characterized by IgA-dominant immune deposits affecting small vessels and often involves the skin, gastrointestinal tract, joints, and kidneys. IgAV is the most common cause of systemic vasculitis in children. The long-term prognosis is dependent on renal involvement: IgAV with nephritis (IgAVN) can progress to renal failure. IgAVN is an inflammatory disease, providing a rationale for the use of corticosteroids. However, data supporting the use of corticosteroids in patients with established IgAVN of any severity remain limited, although most clinicians use them. Even in patients with severe forms of IgAVN, methylprednisolone pulses added to oral corticosteroids appears to improve renal outcomes. Considering the multihit hypothesis for the pathogenesis of IgAVN, involving many other immune agents, there is a strong rationale for the use of other immunosuppressive drugs in patients with IgAVN, including mycophenolic acid, cyclophosphamide, rituximab, calcineurin inhibitors, and complement inhibitors. Thus, these immunosuppressive treatments have also been evaluated in IgAVN, usually in corticosteroid-dependent or corticosteroid-resistant forms and in small retrospective studies. However, their efficacy has not been proven. Thus, the risk of progression to renal failure and the ongoing debate about the best management of IgAVN justifies the interest in investigating and identifying treatments that can potentially preserve renal function in patients with IgAVN. This review reports on the efficacy of the different drugs currently used for the treatment of IgAVN in adults and children.

1 Introduction

Immunoglobulin A (IgA) vasculitis (IgAV), also referred to as Henoch–Schönlein purpura, is characterized by IgA1-dominant immune deposits affecting small vessels (predominantly capillaries, venules, or arterioles); often involves the skin, gastrointestinal tract, joints, and kidneys; and frequently causes arthritis (Table 1) [1]. Glomerulonephritis indistinguishable from IgA nephropathy (IgAN) may occur.

IgAV is the most common cause of systemic vasculitis in children [2]. The long-term prognosis for patients with IgAV depends on the severity of renal involvement. In most series, IgAV nephritis (IgAVN) is diagnosed in about 30% of

Key Points

Data supporting the use of immunosuppressive treatments in children with immunoglobulin A vasculitis with nephritis (IgAVN) remain very limited, and IgAVN treatment remains a matter of debate.

Severe forms of IgAVN should receive methylprednisolone pulses in combination with oral corticosteroids, and steroid treatment must not be delayed.

Long-term follow-up is necessary in patients with IgAVN for whom renal outcomes are uncertain.

If proteinuria persists after a course of corticosteroids, current data suggest that mycophenolate mofetil, rituximab, or tacrolimus should be considered.

✉ Jean-Daniel Delbet
jean-daniel.delbet@aphp.fr

¹ Pediatric Nephrology Unit, Trousseau Hospital, APHP, DMU Origine, Paris, France

² Sorbonne Université, Paris, France

³ Department of Pediatric Nephrology, Armand Trousseau Hospital, 26 Avenue du Docteur Arnold Netter, 75012 Paris, France

Table 1 Immunoglobulin A vasculitis diagnostic criteria according to EULAR/PRINTO/PRES criteria for Henoch–Schönlein purpura [1]

Purpura (mandatory criterion)	Purpura (commonly palpable and in crops) or petechiae, with lower limb predominance and not related to thrombocytopenia and at least one of the four following criteria
Abdominal pain	Diffuse abdominal colicky pain with acute onset assessed by history and physical examination. May include intussusception and gastrointestinal bleeding
Histopathology	Typically leukocytoclastic vasculitis with predominant IgA deposit or proliferative glomerulonephritis with predominant IgA deposit
Arthritis or arthralgias	Arthritis of acute onset defined as joint swelling or joint pain with limitation on motion Arthralgia of acute onset defined as joint pain without joint swelling or limitation on motion
Renal involvement	Proteinuria > 0.3 g/24 h or > 30 mmol/mg of urine albumin/creatinine ratio on a spot morning sample Hematuria or red blood cell casts: > 5 red blood cells/high power field or red blood cells casts in the urinary sediment or ≥ 2+ on dipstick

EULAR European League Against Rheumatism, *IgA* immunoglobulin A, *PRES* Paediatric Rheumatology European Society, *PRINTO* Paediatric Rheumatology International Trials Organisation

patients [3], whereas the proportion of patients with progression to chronic renal failure ranges from 2 to 20% depending on the patient recruitment of the pediatric center [4]. There is a strong association between severe initial clinical and/or histological manifestation and the development of chronic kidney disease (CKD), but patients with subtle biological alterations, such as mild proteinuria or, exceptionally, isolated hematuria at disease onset, can also experience poor outcomes [5]. This makes it difficult to identify easily available biological parameters to establish an individual long-term prognosis.

Corticosteroids and renin–angiotensin system (RAS) blockers are widely used in children presenting with nephrotic-range proteinuria or acute nephritis, and some studies have highlighted a potential beneficial impact of steroid treatment on long-term renal prognosis [6, 7]. However, IgAVN management remains very controversial. Data supporting the use of corticosteroids in children with established nephritis of any severity remain very limited [8], and clinicians have no information to help them determine when prednisone should be started and for how long RAS blocker therapy should be administered before starting corticosteroids [9]. Thus, in the absence of sufficient long-term data, KDIGO (Kidney Disease – Improving Global Outcomes) guidelines recommended IgAVN be treated as isolated IgAN [9], meaning that patients with non-severe forms of IgAVN would first receive a course of RAS blockers without immunosuppressive drugs or steroids, with the risk of undertreating glomerular inflammation. KDIGO guidelines mainly highlighted that IgAVN treatment remains a matter of debate.

Thus, the risk of progression to renal failure and the ongoing debate about the best treatment of IgAVN justifies the investigation into and identification of treatments able to preserve renal function in patients with IgAVN. This review first provides a brief overview of the pathophysiology of IgAVN then reports on the efficacy of the different drugs

currently used for the treatment of IgAVN in adults and children. IgAVN and IgAN are two glomerular nephropathies that share many pathophysiological similarities and the same immune origin [11]; however, clinical research concerning the treatment of IgAN is more abundant than that for IgAVN. For this reason, we also report the most important studies on IgAN treatment, which can be useful for clinicians treating patients with IgAVN.

2 Pathophysiology

IgA1 is the main component of immune deposits in IgAV. In IgAN and IgAVN, O-glycans in the hinge region of human IgA1 are aberrantly glycosylated [10], thereby exposing adjacent N-acetylgalactosamine residues, which can function as neoepitopes. Autoantibodies recognizing these neoepitopes can consequently form immune complexes by binding to galactose-deficient IgA1 (IgG-Gd-IgA1) [11]. IgA is able to bind to multiple receptors. One of these is the transferrin receptor (cluster of differentiation [CD]-71). Overexpression of CD71 by mesangial cells facilitates the deposition of IgG-Gd-IgA1 immune complexes in the mesangium [12]. Renal deposits lead to mesangial cell activation, matrix production, proinflammatory cytokine release, and endocapillary influx of inflammatory cells, and all these processes ultimately lead to sclerosis [13]. Therefore, IgAVN is considered an inflammatory disease, providing the rationale for the use of corticosteroids.

IgAVN pathophysiology is more complicated and involves many other agents, such as FcαRI, which is present as a transmembrane receptor on myeloid cells and in soluble form (sCD89), and its activation results in multiple proinflammatory functions, such as phagocytosis, production of reactive oxygen species, antibody-dependent cellular cytotoxicity (ADCC), or the release of neutrophil extracellular traps [14]. Anti-endothelial cell antibodies (AECA), which

are a heterogeneous group of antibodies directed against antigens on human endothelial cells, likely also play a role in the pathophysiology of IgAVN: IgA from the serum of patients with IgAV binds to human endothelial cells in vitro, supporting the presence of IgA AECA [15]. AECA induce vascular damage in several animal models by endothelial cell activation and ADCC or complement-dependent cytotoxicity [16]. Accumulating evidence also now indicates that complement dysregulation is involved in IgAVN or IgA pathophysiology: biomarkers of alternative and mannan-binding lectin pathway activation are correlated with disease severity [17, 18], and genetic studies have identified IgAN-associated loci involving genes that encode complement regulators [19].

Considering the multihit hypothesis for the pathogenesis of IgAVN [20], there should be a strong rationale for the use of immunosuppressive drugs such as mycophenolic acid, cyclophosphamide, rituximab, calcineurin inhibitors (CNIs), or complement inhibitors in patients with IgAVN.

3 First-Line Treatment of Immunoglobulin A Vasculitis with Nephritis (IgAVN)

3.1 Corticosteroids and Renin–Angiotensin System Blockers

3.1.1 Oral Corticosteroids

As mentioned, corticosteroids are widely used by clinicians to treat IgAVN, despite a lack of clear evidence of their efficacy. The first studies to evaluate the efficacy of corticosteroids in IgAVN were published in the 1970s.

In 1977, Counahan et al. [21] reported the prognosis of 88 children with IgAVN with a median follow-up of 9.9 years. As in the original paper, and in many further studies [5, 22–24], renal outcomes were classified into four categories: (A) normal: normal urine and renal function; (B) minor urinary abnormality: microscopic hematuria or proteinuria < 1 g/day; (C) active renal disease: proteinuria > 1 g/day or hypertension, and glomerular filtration rate (GFR) > 60 mL/min/1.73 m²; (D) renal insufficiency: GFR < 60 mL/min/1.73 m² or death. States A and B were considered to represent good outcomes and states C and D to represent poor outcomes.

In the study by Counahan et al. [21], 38 patients did not receive steroid or immunosuppressive treatment. After a follow-up of more than 6.5 years, 30 of these patients were in remission, three were classified as group B, three as group C, and two as group D. Of the 21 patients who received steroids, 15 were in remission, two were classified as group B, two as group C, and two as group D. No statistical differences were found between treated and untreated patients.

Thus, in the 1970s, researchers faced the same difficulties as current researchers assessing the impact of corticosteroids on the IgAVN course: retrospective studies, small numbers of patients, heterogeneous clinical and histological presentation, and a significant proportion of patients with favorable outcomes without treatment. In other words, progress in the field of IgAVN clinical research has been limited over the last 50 years.

Indeed, we find several examples of similar difficulties in recent studies.

In 2011, Wakaki et al. [25] studied 42 children with IgAVN who presented in a nephrotic state during the early phase of the disease. Of these, 17 patients did not receive immunosuppressive treatment and the remaining 25 patients received corticosteroids. Univariate logistic regression analyses revealed that the duration of the nephrotic state, initial renal insufficiency, and age at onset were significant predictors of long-term outcomes, but types of treatment were not significant predictors [25].

In 2002, a cohort of 250 adult patients with IgAVN was retrospectively analyzed for a median follow-up period of 14.8 years. Of these patients, 110 did not receive treatment and 93 (37%) received steroids alone. Renal outcomes did not differ between those receiving and not receiving steroids. However, proteinuria and creatinine levels were significantly higher at disease onset in the treated group of patients, which makes the results difficult to interpret [26].

Another recent study included 159 children with IgAVN and had a median follow-up of 3 years. Steroid treatment was not associated with favorable renal outcomes, but only 11% of patients did not receive steroid treatment [27].

In retrospective studies, the proportion of patients receiving steroid treatment was almost systematically high, and this is one reason why the impact of corticosteroids remains difficult to establish. It also means that most clinicians used corticosteroids, even though their efficacy remains unproven.

However, several studies highlight a potential beneficial impact of corticosteroids in IgAVN.

A post hoc analysis of one placebo-controlled randomized controlled trial (RCT) showed that IgAVN resolved more rapidly in children treated with prednisone than in those receiving placebo. Indeed, of 36 children, seven (19%) in the prednisone group still had kidney involvement at 6 months compared with 15 of 35 (43%) in the placebo group. However, the trial only provided outcome data at 6 months after randomization. Therefore, it remains unclear whether prednisone treatment reduced the number of patients with persistent IgAVN or only promoted more rapid resolution of kidney disease compared with placebo [6].

In a recent study, exposure to corticosteroids significantly decreased serum levels of Gd-IgA1, which is clearly associated with IgAVN pathophysiology [28]. In another recent French study, corticosteroid use seemed to be associated

with good renal outcomes in children with IgAVN and International Study of Kidney Disease in Children (ISKDC) class 2 histology (without crescents) [29].

Many studies have evaluated the impact of corticosteroids in IgAN, and their efficacy appears to be established. In the 1980s, three RCTs showed clear evidence of benefits from the long-term use of steroids (at least 6 months) in reducing proteinuria and disease progression with loss of renal function [30–32].

A review of the Cochrane Database on immunosuppressive agents in the treatment of IgAN analyzed 32 studies including 1781 patients and showed that corticosteroids significantly reduced the risks of progression to end-stage renal disease and doubling of serum creatinine, also reducing proteinuria [33]. However, it is important to note the debate concerning the use of corticosteroids in IgAN. For example, the authors of the STOP-IgAN RCT concluded that immunosuppressive therapy in addition to optimal supportive care would not provide substantial kidney-related benefits in Caucasian patients with high-risk IgAN [34].

3.1.1.1 Steroids to Prevent IgAVN The use of corticosteroids to prevent nephritis in patients with IgAV has been also analyzed [35]. In 2013, a total of 352 children were randomized to receive corticosteroids 2 mg/kg/day for a week followed by 1 mg/kg/day for another week or placebo for 14 days and were evaluated for kidney outcomes after 12 months. No evidence was found to show that early steroid treatment reduced the risk of proteinuria during the ensuing 12 months [35]. Thus, we conclude that corticosteroids do not prevent nephritis in patients with IgAV.

3.1.2 Methylprednisolone Pulses

Methylprednisolone pulses added to oral corticosteroids are widely used in the treatment of IgAVN, especially in severe forms.

In 1998, Niaudet and Habib conducted a particularly important study showing the possible benefits of methylprednisolone pulses [7]. This study reported that methylprednisolone pulses considerably improved renal outcomes in 38 children with severe forms of IgAVN. Moreover, renal biopsies performed before and after methylprednisolone pulses showed a substantial post-treatment decrease of active histological lesions (such as endocapillary or extracapillary proliferation) [7]. However, there was no control group, and a spontaneous favorable course of IgAVN could not be excluded. However, in comparison with a historical group for patients not treated with methylprednisolone pulses, there was a significant difference in long-term outcomes.

In 2003, Kawasaki et al. [37] presented renal outcomes in 56 patients with IgAVN with crescents treated with methylprednisolone pulses associated with urokinase. Proteinuria

decreased 6 months after treatment onset, as did active histological lesions on control biopsies [37].

In 2008, Katafuchi et al. [38] demonstrated that methylprednisolone pulses had greater efficacy than oral corticosteroid therapy in patients with clinically and histologically severe presentation.

It is important to note that all patients in these studies had severe histological features with extracapillary proliferation. Like patients with anti-neutrophilic cytoplasmic autoantibody (ANCA) vasculitis with glomerulonephritis, patients with IgA glomerular deposits and extracapillary proliferation should benefit from methylprednisolone pulses, even though the severity of progression in these two entities is not comparable. These data are presented Table 2.

3.1.2.1 Renin–Angiotensin System Blockers Similarly, there is no clear evidence of the efficacy of RAS blockade in IgAVN. However, there is much data to suggest that angiotensin-converting enzyme inhibitors (ACEI)/angiotensin-receptor blockers (ARBs) should be used in IgAVN and as soon as possible.

A recent study evaluated renal outcomes in 142 children with IgAVN. They reported that treatment with ACEI/ARBs was mainly linked to less proteinuria if it was started early after renal disease onset. There was a significant positive correlation between time from disease onset to start of ACEI/ARB and proteinuria at 6 months ($p = 0.03$), 12 months ($p = 0.07$), and 9–10 years ($p = 0.01$) [39].

Ninchoji et al. [40] reported the results of a series of 31 pediatric patients with moderately severe Henoch–Schönlein purpura nephritis (histological grades I–III and serum albumin > 2.5 g/dL) treated only with ACEI/ARBs. After a follow-up of 3.75 years, all patients were in remission [40]. In IgAN, prospective randomized trials have shown that the use of ACEI/ARBs improves renal outcomes in adults and children [41–43].

4 Other Immunosuppressive Treatments

Many other immunosuppressive treatments have been evaluated in IgAVN, mainly in forms dependent on or resistant to corticosteroids (Table 2).

4.1 Calcineurin Inhibitors

Some studies have reported on the effects of CNIs in severe and steroid-resistant forms of IgAVN.

The efficacy of cyclosporine was initially reported in the early 2000s in a small cohort of children. In 2003, Ronkainen et al. [44] showed that IgAVN remission was achieved with cyclosporine in four of seven children after failure of steroid therapy, but these children developed

Table 2 Treatment of immunoglobulin A vasculitis with nephritis: primary studies

Study	Patients (n)	Study design	Therapy	Follow-up (median)	Renal outcome
CS or methylprednisolone pulses					
Pillebout et al. [26]	250 adults	Retrospective study	No treatment (n = 110); CS alone (n = 93)	14.8 years	No difference between the two groups
Wakaki et al. [25]	42 children in a nephrotic state	Retrospective study	No treatment (n = 17); CS alone (n = 25)	6.2 years	No difference between the two groups
Delbet et al. [27]	159 children	Retrospective study	11% of patients did not receive steroid treatment	3 years	Steroid treatment was not associated with favorable renal outcomes
Ronkainen et al. [6]	171 children	Post hoc analysis of placebo-controlled trials	Oral CS 1 mg/kg/day BID for 14 days, followed by a weaning dose of 0.5 mg/kg/day for 1 week, then 0.5 mg/kg/day OD on alternate days for an additional week	6 months	IgAVN resolved more rapidly in children treated with prednisone vs. placebo
Niaudet and Habib [7]	38 children with nephrotic syndrome and/or ≥ 50% crescentic glomeruli	Prospective study	Methylprednisolone pulses for 3 days followed by oral CS for 3.5 months	1–16 years	27 children recovered; seven persistent nephritis; four end-stage renal failure
Other immunosuppressive treatments					
Jauhola et al. [46]	24 children with nephrotic-range proteinuria or crescents in kidney biopsy	Randomized controlled prospective study	CsA, initial dose 5 mg/kg/day for 1.2 years or IV methylprednisolone pulses of 30 mg/kg followed by prednisone 60 mg/m ² /day for 3 months	6 years	All CsA patients experienced remission within 3 months; remission not experienced in 43% of patients treated with methylprednisolone pulses
Zhang et al. [49]	20 children	Prospective observational study	Tacrolimus treatment initiated at 0.05–0.1 mg/kg/day BID for ≥ 6 months	6 months	12 patients experienced CR and eight experienced PR at the end of the 6-month treatment
Han et al. [61]	95 adults	Retrospective study	Three groups: MMF group (n = 33) received MMF 1.0–1.5 g/day combined with prednisone 0.4–0.5 mg/(kg·day); CS group (n = 31) received full-dose prednisone 0.8–1.0 mg/kg/day; and a control group	28 months	Remission rates in the MMF, CS, and control groups were, respectively, 72.7%, 71.0%, and 48.4% at 6 months and 72.7%, 64.5%, and 45.2% at the end of follow-up
Maritati et al. [67]	22 adults	Retrospective study	RTX either for relapsing disease or because they had contraindications to CS	24 months	20 patients (90.9%) experienced remission; seven of those (35%) had subsequent relapse of disease

BID twice daily, *CR* complete remission, *CS* corticosteroids, *CsA* cyclosporine, *IgAVN* immunoglobulin A vasculitis with nephrology, *IV* intravenous, *MMF* mycophenolate mofetil, *OD* once daily, *PR* partial remission, *RTX* rituximab

cyclosporine dependency. In 2005, Shin et al. [45] reported on the use of combination cyclosporine and corticosteroid therapy in eight children with IgAVN. Cyclosporine was also effective in improving histological lesions and proteinuria in these patients [45].

In 2011, a randomized trial including 24 pediatric patients with IgAVN with a mean age of 9.4 years compared a 12-month treatment regimen with cyclosporine versus three methylprednisolone pulses followed by corticosteroids [46]. Eleven patients were randomized to receive cyclosporine at an initial dose of 5 mg/kg/day with a mean duration of treatment of 1.2 years, and 13 patients received methylprednisolone pulses in three doses of 30 mg/kg followed by prednisone 60 mg/m²/day over 1 month, after which the medication was gradually tapered over 3 months. The mean follow-up was 6 years. All cyclosporine-treated patients experienced remission within 3 months; however, in the methylprednisolone pulses group, remission was not achieved in 43% of patients, and other immunosuppressive treatments had to be started after a mean of 1.6 years. The difference in treatment was statistically different at 1 and 3 months ($p = 0.04$ and 0.016 , respectively). Thus, the impact of cyclosporine on renal outcomes seemed very promising.

IgAVN data in adult patients are scarce. Kalliakmani et al. [47] reported IgAVN remission induced by cyclosporine in five patients with nephrotic-range proteinuria.

Tacrolimus has also been used in children with IgAVN [48] and should be considered as a promising steroid-sparing agent for the treatment of severe IgAVN. In a pilot study with 20 pediatric patients with IgAVN, Zhang et al. [49] reported that 12 patients reached complete remission (CR) and eight reached partial remission at the end of 6 months of tacrolimus treatment.

In IgAN, a recent meta-analysis including 370 patients assessed the effects of CNIs: five studies showed that CR of nephritis occurred more frequently in the CNIs group than in the control group in a total of 225 patients (risk ratio 2.51; 95% confidence interval [CI] 1.25–5.04). Proteinuria was evaluated in 169 patients, with 24-h proteinuria significantly lower in patients using CNIs than in the control groups (weighted mean difference -0.46 g/day, 95% CI -0.55 to -0.24) at the end of treatment or during follow-up.

Thus, CNIs might provide renal protection in patients with IgAN. However, CNI therapy was also associated with an increased risk of gastrointestinal and neurological symptoms and hirsutism [50].

In summary, CNIs should be considered a promising agent for the treatment of severe IgAVN.

4.2 Cyclophosphamide

Cyclophosphamide was the subject of an RCT in which 28 treated children with IgAVN were compared with a control

group of 28 children with IgAVN receiving only symptomatic treatment. After 7 years of follow-up, no differences between the two groups were shown in terms of CR or partial renal remission or in the frequency of end-stage renal disease [51].

In adults, Pillebout et al. [52] compared high-dose prednisone alone versus high-dose prednisone plus cyclophosphamide (six pulses of 0.6 g/mq) in 54 patients with IgAVN. No difference was found between the two groups in remission rate at 6 months or in renal outcomes and deaths at 12 months.

Audemard-Verger et al. [53] analyzed the outcomes of 80 patients treated with corticosteroids alone and 30 treated with corticosteroids and cyclophosphamide and found no differences in the frequency of partial or complete responses to therapy within the first 12 months of follow-up.

These data are not in favor of the use of cyclophosphamide in IgAVN.

4.3 Azathioprine

Azathioprine has mainly been used in combination with corticosteroid therapy in uncontrolled small cohorts of pediatric patients. This combination has been shown to be effective in the treatment of severe forms of IgAVN [54]. Other retrospective studies seemed to confirm these results [55, 56]. In IgAN, a meta-analysis including 325 patients was published in 2016. Patients were divided into three groups according to treatment: no treatment, corticosteroids, and corticosteroids plus azathioprine. No additional benefit was reported in patients receiving additional azathioprine treatment [57].

4.4 Mycophenolate Mofetil

Recently, the efficacy of mycophenolate mofetil (MMF) in IgAVN was reported in two case series including, respectively, six and 12 children initially treated with steroids. MMF was effective in inducing remission and preventing relapses [58, 59]. In 2012, Ren et al. [60] retrospectively compared 27 adult patients with IgAVN treated with MMF and low-dose prednisone (0.4–0.5 mg/kg/day) versus 26 patients treated with higher doses of prednisone alone (0.8–1.0 mg/kg/day). After a median of 28.2 months, 77.8% of patients included in the MMF group achieved remission, the same proportion as those receiving full-dose corticosteroid therapy [60]. In 2015, Han et al. [61] reported the outcomes of 95 patients with IgAVN. Patients were retrospectively grouped according to treatment: MMF combined with low-dose prednisone, full-dose prednisone, or ACEI alone. Patients treated with MMF had a higher remission rate than both the control group and those receiving full-dose prednisone; those receiving MMF plus low-dose prednisone had

the same remission rate, fewer adverse effects, and a tendency to fewer relapses at the end of the follow-up.

In IgAN, a recent meta-analysis reported pooled results by combining eight studies and suggested that patients with IgAN in the group receiving mycophenolic acid had a higher remission rate than those in the control group [62].

These data are very interesting for the efficacy of MMF in IgAVN, albeit they are from retrospective studies. Moreover, MMF is frequently used in nephrology, and the tolerance profile is rather good. Thus, we consider that MMF should be proposed in patients with IgAVN, particularly if proteinuria persists after an initial steroid course and despite antiproteinuric treatment.

4.5 Rituximab

Rituximab is an anti-CD20 monoclonal antibody that induces B-cell depletion and has become a standard treatment in many autoimmune disease with renal involvement [63]. A few reports suggested that rituximab could be efficient in IgAVN in both pediatric and adult patients [64–66]. In 2018, a prospective observational study reported outcomes in 22 adult patients with IgAVN treated with rituximab for relapsing or refractory disease or because they had contraindications to standard-dose corticosteroids or conventional immunosuppressive therapy. The authors observed a remission rate of 91% within 6 months of therapy [67].

In IgAN, a trial including 34 adults with IgAVN compared rituximab with supportive care and failed to show an improvement in estimated GFR (eGFR), proteinuria, IgG-Gd-IgA1, or anti-IgG-Gd-IgA1 antibody levels after 12 months [68].

Data are still scarce, but rituximab should be considered in patients with IgAVN, especially if other oral immunosuppressive treatments have failed to induce IgAVN remission.

4.6 Plasmapheresis

Several studies have reported the use of plasmapheresis in small groups of patients with IgAVN and very severe nephritis. In three retrospective studies, plasmapheresis did not prevent progression to end-stage kidney disease in a substantial proportion of children (28–50%) [69–71].

However, another retrospective study including 14 children with severe IgAVN showed that only one child required a kidney transplant at the end of 4 years of follow-up, and the other patients recovered without renal failure [72].

Augusto et al. [73] reported the outcomes of 11 adult patients with severe IgAVN treated with corticosteroids and 12 plasma exchange sessions. The authors demonstrated a rapid improvement in proteinuria and kidney function at 12 months, and no relapse of IgAVN was noted after a median follow-up of 6 years [73].

Plasmapheresis can be used as rescue therapy in case of rapid renal function degradation or persistent nephrotic syndrome.

5 Dapsone

Dapsone in the treatment of IgAV was reported for the first time in 1983 [74] and has since been used in several other cases of IgAV. Persistent rash is clearly the main indication for dapsone in IgAV, and outcomes for skin involvement seem interesting, although no controlled study has been conducted [75].

Outcomes with the use of dapsone in IgAVN are more limited. In a recent review, proteinuria was identified in six of 26 patients with IgAV treated with dapsone; unfortunately, as in previous case series [75], renal presentations and outcomes were not clearly exposed [74].

Hence, to date, although good outcomes have sometimes been described in these patients [75], there is no strong evidence of a positive effect with dapsone in IgAVN.

However, in our view, dapsone should be considered for IgAVN treatment as it has antioxidant scavenger effects and may suppress the production of toxic free radicals by neutrophils. It also decreases the synthesis of IgA antibodies [76]. All these phenomena are essential in the pathophysiology of IgAVN.

6 Complement Activation in Immunoglobulin A Nephropathy

Activation of both the alternative pathway and the lectin pathway have been shown in IgAN [77]. Mannan-binding lectin-associated serine proteinase 2 (MASP-2) is essential for lectin pathway activation so is a potential target for future drugs. Narsoplimab is a humanized monoclonal antibody and a MASP-2 inhibitor that has been designed to treat pathologies mediated by lectin pathway activation of the complement system. A phase II study showed that narsoplimab was well-tolerated. It appears to result in a clinically significant proteinuria reduction and stability of renal function assessed with eGFR in high-risk patients with IgAN [24].

7 Prognostic Factors

Prognostic factors remain a matter of debate in IgAVN. Obviously, severe clinical (renal insufficiency or nephrotic syndrome) or histological (diffuse extracapillary proliferation) presentation have been correlated with poor renal outcomes, and these patients should benefit from more

aggressive treatment. Since the 1970s, authors have highlighted that poor outcomes are correlated with a clinical presentation of combined acute nephritis and nephrotic syndrome and a high proportion of crescents in the kidney biopsy [21]. More recently, a German study associated nephrotic syndrome with a primary outcome of kidney failure (eGFR < 40 mL/min/1.73 m²) [71], and a Swedish study showed that decreased eGFR, hypertension, and the presence of proteinuria at the time of biopsy were significantly associated with the outcome of kidney failure [78].

However, as usual, other studies found no correlation between renal outcomes and initial presentation. For example, in an Italian cohort of 83 pediatric and 136 adult patients, decline in eGFR was not correlated with initial clinical features, including CKD, proteinuria, and hypertension [79], and a recent study of a French pediatric cohort reported that only chronic histological lesions were correlated with the absence of nephritis remission in multivariate analysis [27].

Moreover, in pediatric patients with IgAVN with initial good histological presentation (without crescents), a low initial proteinuria was associated with poor renal outcomes, meaning that individual prognoses for patients with IgAVN remain difficult to establish [29].

8 Recommendations and Conclusion

Given the lack of evidence, clinicians must remain careful when treating IgAVN. Long-term follow-up is necessary in these patients, for whom renal outcomes are uncertain, and the side effects of treatments must be regularly checked. Obviously, initial treatment will depend on initial clinical and/or histological presentation.

We recommend that patients with severe forms of IgAVN (nephrotic syndrome, extracapillary proliferation on renal biopsy) receive methylprednisolone pulses in combination with oral corticosteroids [7]. If treatment fails or renal function deteriorates rapidly, plasma exchange should be considered.

We recommend that patients with moderate forms of IgAVN (nephrotic proteinuria without nephrotic syndrome or renal failure, endocapillary proliferation without crescents) receive oral corticosteroids for 4–6 months.

In patients with mild IgAVN (proteinuria < 100 mg/mmol, isolated hematuria), it is possible that immunosuppressive treatments may not be necessary, but these patients must be closely monitored and immunosuppressive treatment quickly proposed if proteinuria persists or increases. In all these cases, RAS blockers should be used at disease onset [39].

If proteinuria persists after a corticosteroid course, current data suggest that MMF seems to be the best

immunosuppressive treatment in IgAVN. Of course, rituximab or tacrolimus could also be considered.

Unlike recommendations in IgAN, methylprednisolone pulses in combination with oral corticosteroids for patients with severe initial presentation, or oral corticosteroids alone for those with moderate initial presentation, must not be delayed. Immunosuppressive treatment should be started as soon as diagnosis is made, even in non-rapidly progressive nephritis. Indeed, it has been reported that a longer delay between nephritis and kidney biopsy was related to a higher proportion of chronic lesions [28,84] and that chronic lesions were insensitive to immunosuppressive therapy [8]. Therefore, it may be dangerous to delay steroid treatment in these patients.

It is important to note that, according to KDIGO guidelines, in patients with persistent proteinuria without renal failure (which is a large proportion of patients with IgAVN), no corticosteroid is recommended, which increases the risk of acute lesions transforming into chronic lesions.

European consensus-based recommendations for IgAVN treatment were recently published. These recommendations supported the use of corticosteroids without delay. Corticosteroids remain the first-line agent for mild IgAVN (normal eGFR and proteinuria < 250 mg/mmol) or moderate IgAVN (< 50% crescents on biopsy, impaired eGFR < 80 mL/min/1.73 m², proteinuria > 250 mg/mmol). The addition of azathioprine, cyclosporine, MMF, or intravenous cyclophosphamide may be considered as second-line agents. Severe IgAVN (> 50% crescents on biopsy, impaired eGFR < 80 mL/min/1.73 m², proteinuria > 250 mg/mmol) should be treated with methylprednisolone or cyclophosphamide pulses and followed by a maintenance phase using lower-dose corticosteroids and azathioprine or MMF [85].

Unfortunately, all these recommendations were based on expert opinion rather than prospective controlled studies. The debate remains open.

Declarations

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